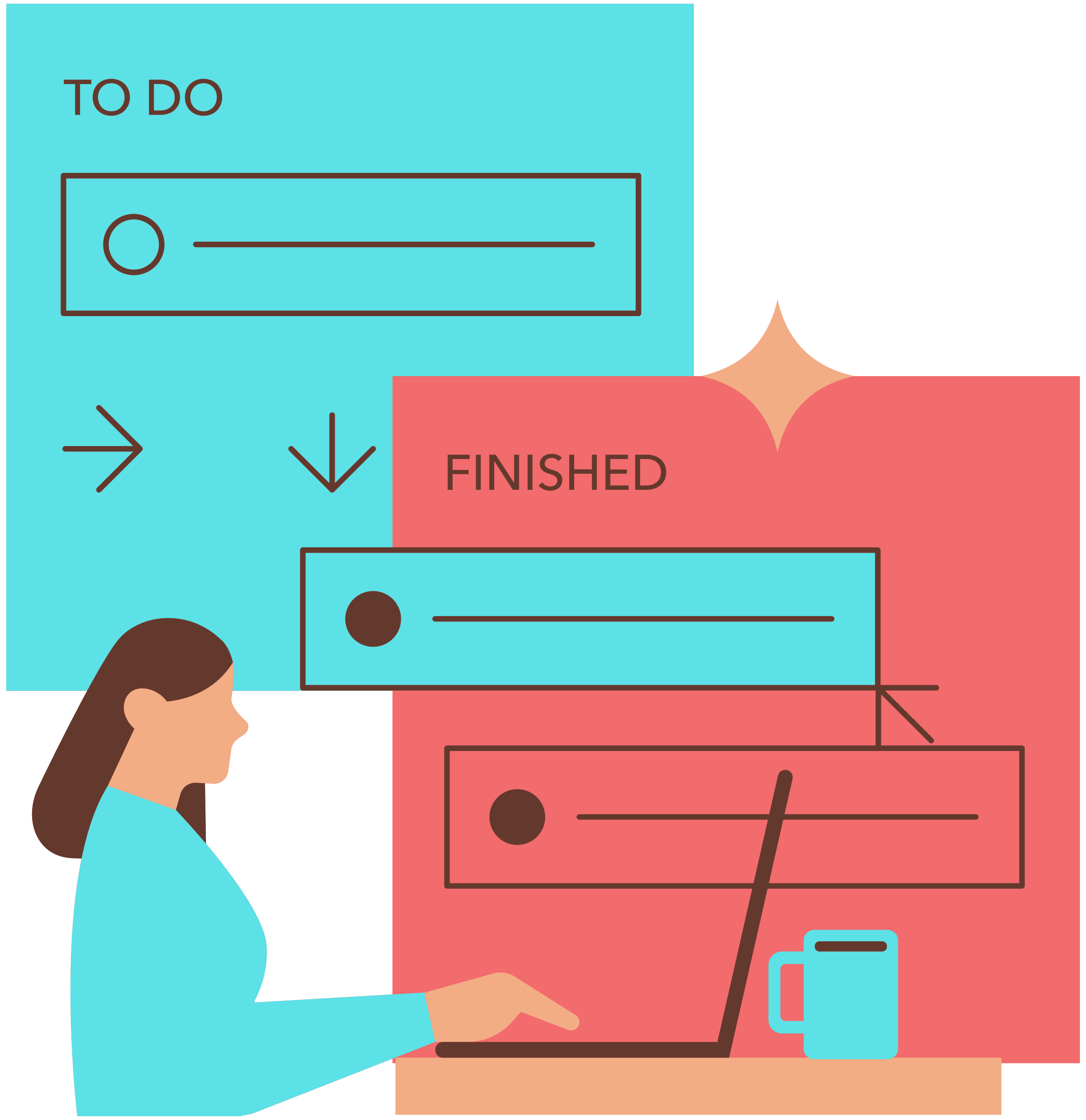


Presentation On

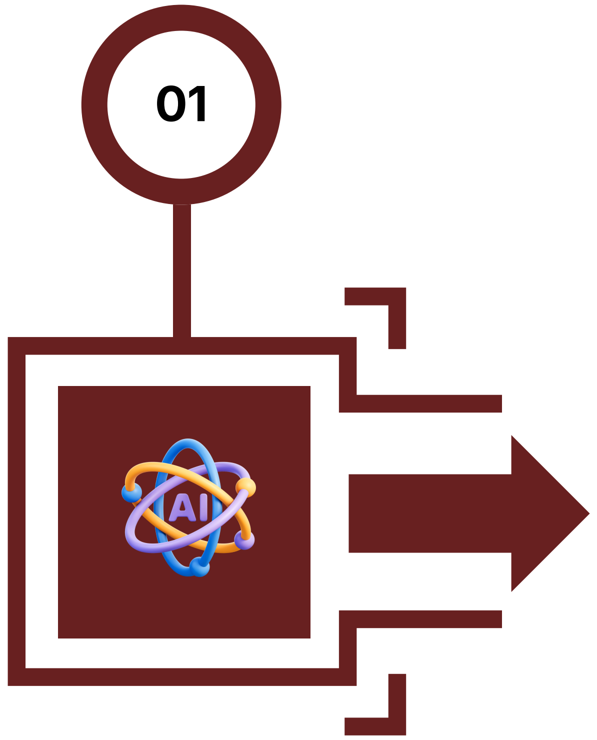
Digital Farming Assistant By Alpha Matrix

A smart farming solution using AI to assist farmers.

meet the group.

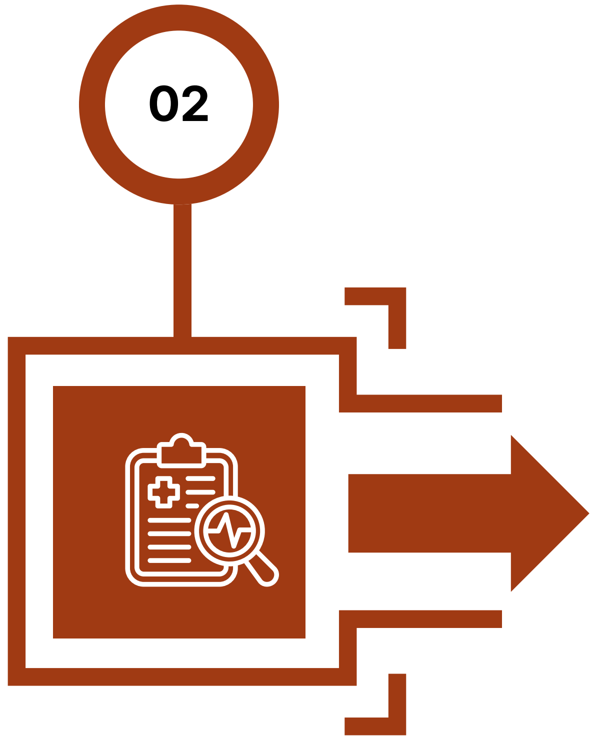
1. MD. Afridi
   ID: 2022-3-60-080
2. Kazi Mazharul Islam Mridul
   ID:: 2023-2-60-098
3. Md Arman Hossain
   ID: 2022-2-60-043
4. Israt jahan tasnim
   ID: 2022-3-60-299
5. Tasnim Jabir
   ID: 2022-3-60-083

Project Overview



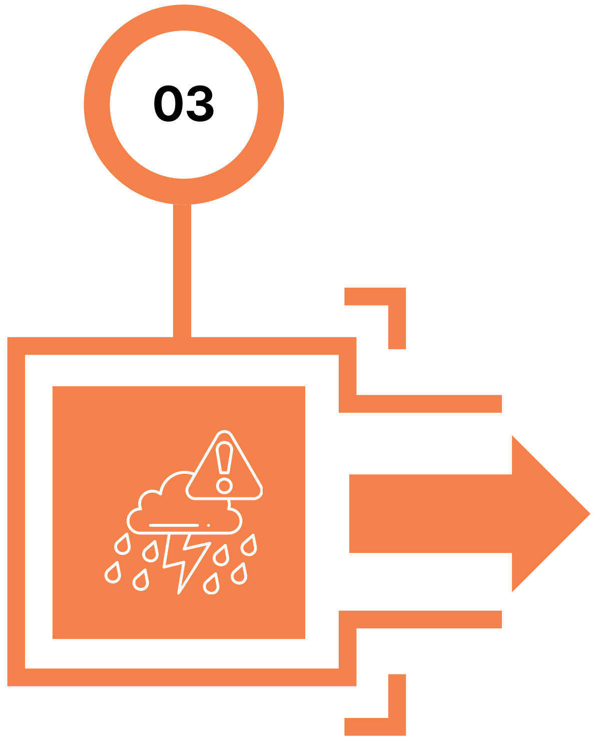
Smart AI Solution

Boosts farming efficiency and productivity.



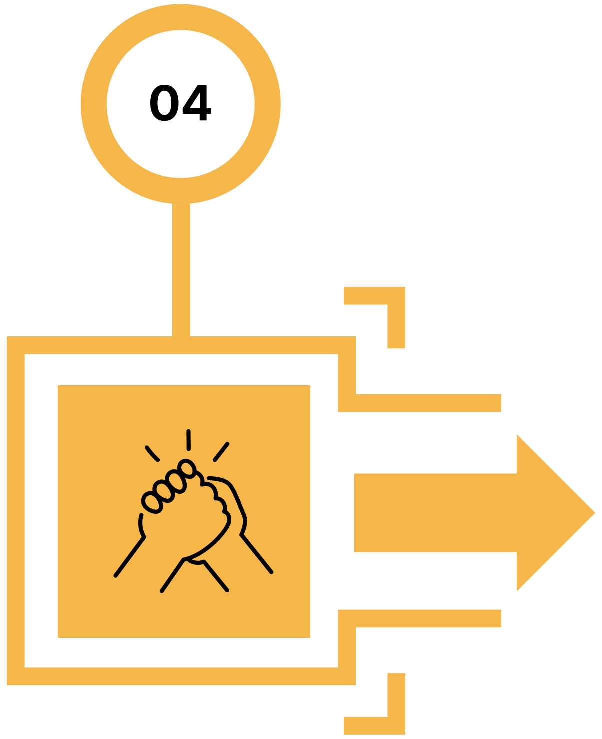
Crop Health Diagnostics

AI powered disease detection and treatment guidance



Weather Alerts & Planning

Real time forecasts and disaster alerts to protect crops.



Community Support

A forum for farmers to share knowledge and solve problems

User Story & Stakeholders

A farmer, Rafiq (Primary Stakeholder - End User), logs into the Digital Farming Assistant using his phone number and preferred language. His profile, including crop history and saved data, helps the system personalize his experience.

As Rafiq works in his field, he captures an image of a diseased crop. Due to low connectivity (Constraint), the image is stored in an offline queue. Once uploaded, the AI Assistant (System Component) analyzes the image and detects a possible disease.

The system generates a diagnosis and recommendations, which are later reviewed and verified by an Agronomist (Stakeholder - Agricultural Expert). This ensures reliability and builds trust for farmers.

The system generates a diagnosis and recommendations, which are later reviewed and verified by an Agronomist (Stakeholder – Agricultural Expert). This ensures reliability and builds trust for farmers.

User Story & Stakeholders

The Project Team / Developers (Stakeholder – Internal) ensure smooth system functionality, including offline support and AI accuracy. The Project Manager oversees execution, while the Project Sponsor (Stakeholder – Authority) monitors overall success

Rafiq can also interact in the Community Forum, where other Farmers (Stakeholders – User Community) and Agricultural Experts share knowledge, answer questions, and provide guidance.

Government Agricultural Officers (Stakeholder – Regulatory Authority) ensure that farmer data is protected and the system follows compliance policies.

By the end of the day, the Digital Farming Assistant connects multiple stakeholders—Farmers, Experts, Developers, Data Providers, and Authorities—into one ecosystem that supports smart and efficient farming.

Key Features

01 Login System

02 Crop Health Checker

03 Weather Forecast

04 Crop Library

05 Community Forum

Login System

Sub Features

- **Name**
- **Phone Number**
- **Preferred languages**

Core Function

- Simple authentication without email, password, or OTP
- Users can choose their preferred language

Crop Health Checker

Sub Features

- **Image Capture & Upload**
- **Diagnosis Summary**
- **Offline Image Queue**
- **Diagnosis History**

Core Function

- Analyzes crop images via ML to detect rot, pests, or diseases.
- Provides specific treatment and care recommendations.
- Supports offline image queuing and access to diagnosis history.

Weather Forecast

Sub Features

- **Daily and weekly forecast view**
- **Location-Based Forecast**
- **Farm Planning advisory**

Core Function

- Displays real-time and weekly forecasts via Open Weather Map API.
- Sends push notifications for severe weather like heavy rainfall or storms.
- Offers irrigation and harvesting advice based on upcoming weather

Crop Library

Sub Features

- **Filter-Based Search**
- **Crop Profiles**
- **Offline Access**

Core Function

- Searchable database with filters for seasons and soil types.
- Guidelines for planting, soil requirements, fertilizers, and pest control.
- Enables offline access by saving most viewed crop profiles.

Community Forum

Sub Features

- **Post with image**
- **Comment**
- **Like**
- **Search questions by keyword**

Core Function

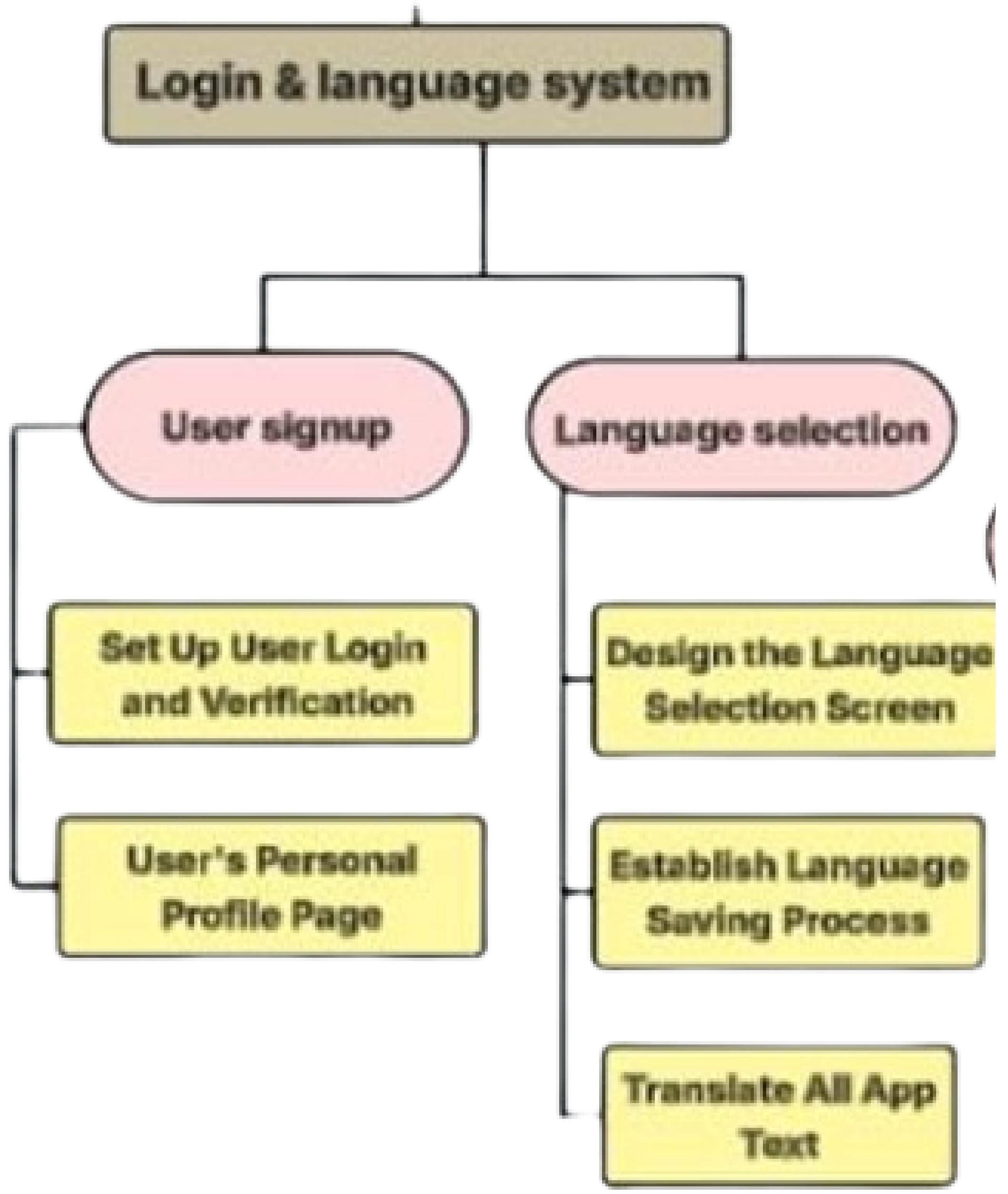
- Enables offline access by saving most viewed crop profiles.
- Comment and like system to highlight useful feedback.
- Keyword search to easily find existing problems and solutions.

WBS Overview

(Login and Language System)

User Signup: Setting up user login, verification, and personal profile pages.

Language Selection: Designing the selection screen, saving preferences, and translating app text



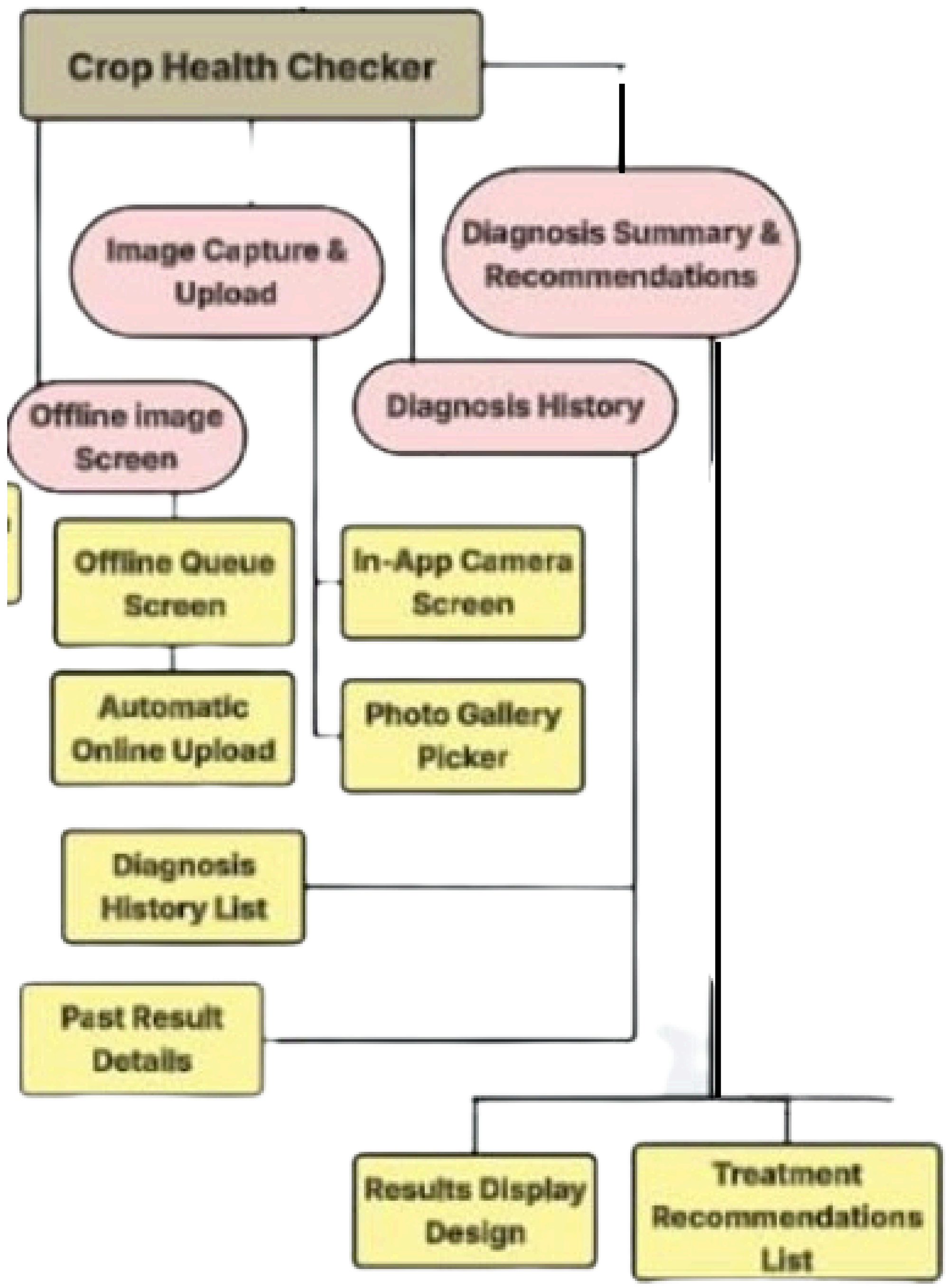
WBS Overview
(Crop Health Checker)

Image Capture & Upload: Developing the In-App Camera Screen and Photo Gallery Picker.

Offline Image Screen: Designing the Offline Queue Screen and Automatic Online Upload feature.

Diagnosis Summary & Recommendations: Creating the Results Display Design and Treatment Recommendations List.

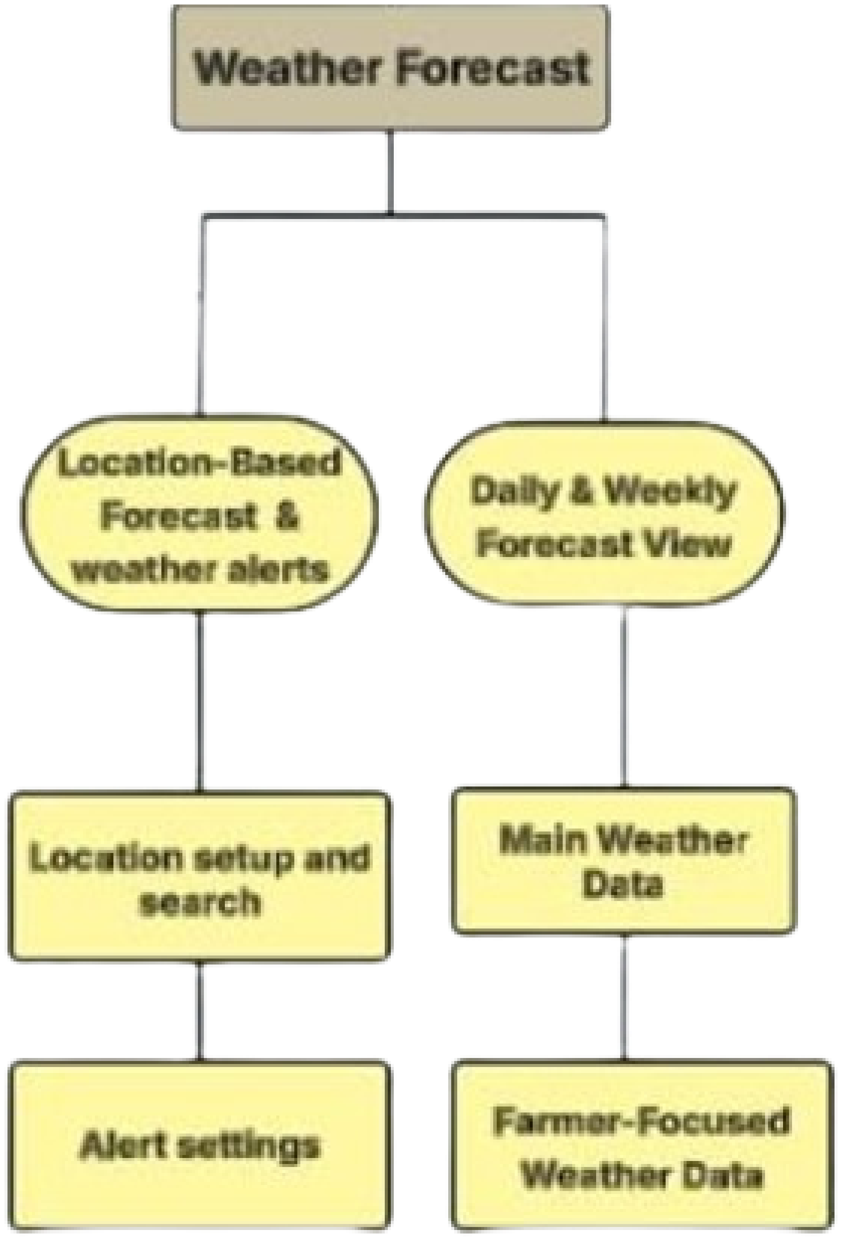
Diagnosis History: Building the Diagnosis History List and Past Result Details view.



WBS Overview
(Weather Forecast)

Location & Alerts: Setting up location-based search and configuring weather alert settings.

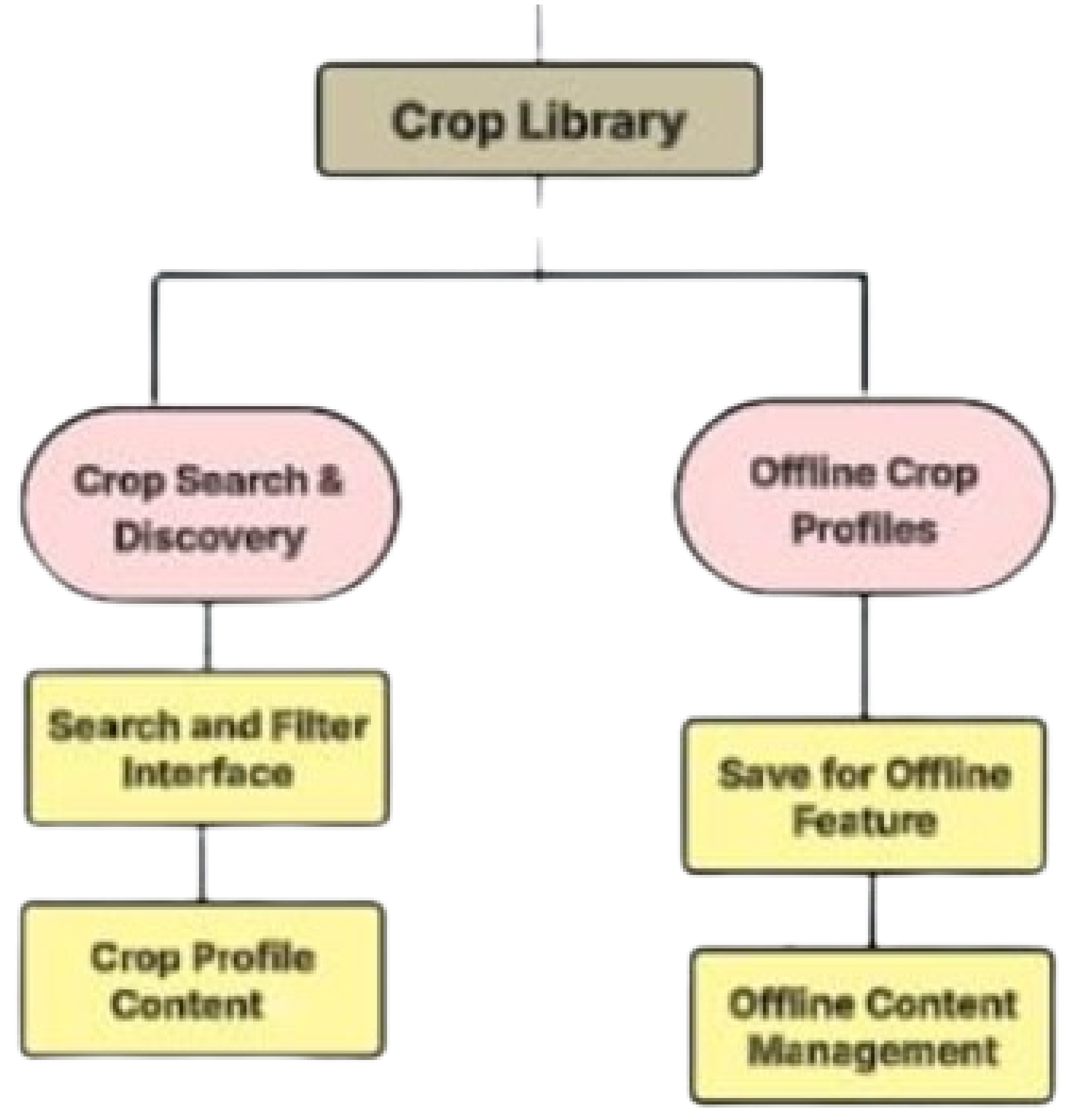
Forecast Views: Integrating main weather data and farmer-focused daily/weekly forecasts.



WBS Overview
(Crop Library)

Search & Discovery: Building interfaces for searching, filtering, and displaying crop profiles.

Offline Access: Implementing features to save and manage crop content for offline use.

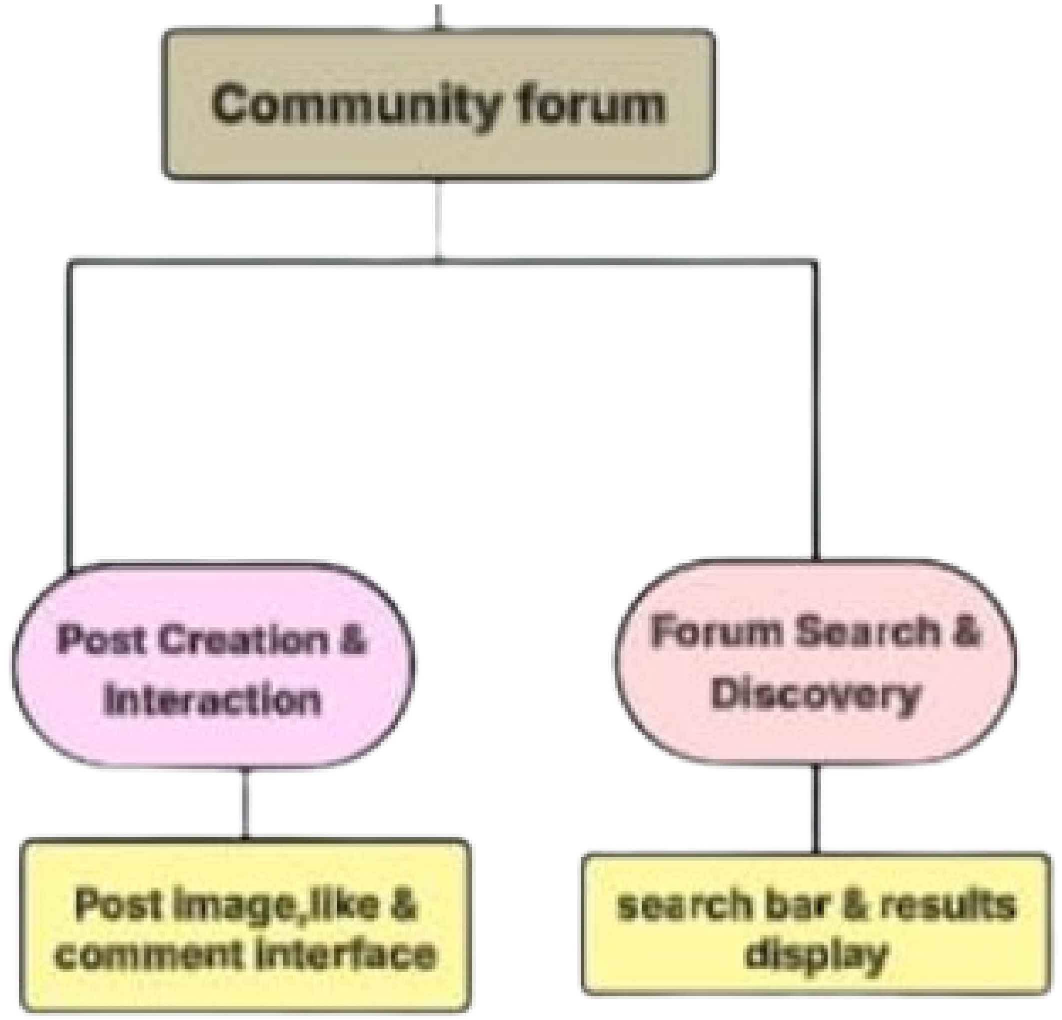


WBS Overview

(Community Forum)

Post & Interaction: Creating user interfaces for posting images, commenting, and liking.

Search & Discovery: Implementing a search bar and results display to find forum discussions.



Product Backlog

Module	ID	User Stories	Initial	Sprint#	Total	S1 done	S1 rem	S2 done	S2 rem	S3 done	S3 rem	S4 done	S4 rem
User Management	1.1	user, create account using name & phone number	14	1	60	14	0	0	0	0	0	0	0
User Management	1.2	user, login using phone number	8	1	60	8	0	0	0	0	0	0	0
User Management	1.3	user, select preferred language	6	1	60	6	0	0	0	0	0	0	0
User Management	1.4	user, view profile & activity history	12	1	60	12	0	0	0	0	0	0	0
User Management	1.5	user, enable offline access	10	1	60	10	0	0	0	0	0	0	0
Crop Health Checker	2.1	user, capture crop image	12	2	85	0	12	8	4	4	0	0	0
Crop Health Checker	2.2	user, upload image (online/offline)	16	2	85	0	16	10	6	6	0	0	0
Crop Health Checker	2.3	system, offline image queue handling	18	2	85	0	18	12	6	6	0	0	0
Crop Health Checker	2.4	system, AI-based image analysis	30	3	95	0	30	0	30	15	15	15	0
Crop Health Checker	2.5	system, show diagnosis & recommendations	18	3	95	0	18	0	18	10	8	8	0
Crop Health Checker	2.6	user, view diagnosis history	10	3	95	0	10	0	10	6	4	4	0
Weather Forecast	3.1	user, location-based weather data	14	2	70	0	14	9	5	5	0	0	0
Weather Forecast	3.2	user, daily & weekly forecast view	12	2	70	0	12	8	4	4	0	0	0
Weather Forecast	3.3	system, weather alerts generation	10	3	70	0	10	0	10	6	4	4	0
Weather Forecast	3.4	system, send notifications	10	3	70	0	10	0	10	6	4	4	0
Weather Forecast	3.5	system, farming advice based on weather	16	4	70	0	16	0	16	0	16	9	7
Crop Library	4.1	user, search crops by filter	14	3	65	0	14	0	14	8	6	6	0
Crop Library	4.2	user, view crop profiles	18	3	65	0	18	0	18	10	8	8	0
Crop Library	4.3	user, save crops offline	10	4	65	0	10	0	10	0	10	6	4
Crop Library	4.4	system, manage crop data	14	4	65	0	14	0	14	0	14	8	6

Sprint Wise Expenditure

Sprint 1				
Person	**Level**	**Weight**	**Allocation**	**Cost**
MD. Afridi	Scrum Master	1.5	100.00%	1.5
Kazi Mazharul Islam Mridul	Junior Developer	0.9	60.00%	0.54
Md Arman Hossain	Senior Developer	1.2	100.00%	1.2
Israt Jahan Tasnim	Junior Developer	0.9	90.00%	0.81
Tasnim Jabir	Senior Developer	1.2	100.00%	1.2
		5.7	450.00%	5.25

Sprint 2				
Person	**Level**	**Weight**	**Allocation**	**Cost**
MD. Afridi	Scrum Master	1.5	100.00%	1.5
Kazi Mazharul Islam	Junior Developer	0.9	100.00%	0.9
Md Arman Hossain	Senior Developer	1.2	80.00%	0.96
Israt Jahan Tasnim	Junior Developer	0.9	100.00%	0.9
Tasnim Jabir	Senior Developer	1.2	100.00%	1.2
		5.7	480.00%	5.46

Sprint 3				
Person	**Level**	**Weight**	**Allocation**	**Cost**
MD. Afridi	Scrum Master	1.5	75.00%	1.125
Kazi Mazharul Islam Mridul	Junior Developer	0.9	100.00%	0.9
Md Arman Hossain	Senior Developer	1.2	95.00%	1.14
Israt Jahan Tasnim	Junior Developer	0.9	80.00%	0.72
Tasnim Jabir	Senior Developer	1.2	60.00%	0.72
		5.7	410.00%	4.605

Sprint Wise Expenditure

	Sprint 4			
Person	Level	Weight	Allocation	Cost
MD. Afridi	Scrum Master	1.5	95.00%	1.425
Kazi Mazharul Islam Mridul	Junior Developer	0.9	100.00%	0.9
Md Arman Hossain	Senior Developer	1.2	100.00%	1.2
Israt Jahan Tasnim	Junior Developer	0.9	100.00%	0.9
Tasnim Jabir	Senior Developer	1.2	50.00%	0.6
		5.7	445.00%	5.025

Metric	Sprint 1	Sprint 2	Sprint 3	Sprint 4	
Cost in Ma	5.25	5.46	4.605	5.025	
Cost in Man Days	26.25	27.3	23.025	25.125	
Actual Cos	5.25	10.71	15.315	20.34	
Earned Value	14.28571	13.4286	24.5714	33.42857143	
Cumulative EV	14.28571	27.7143	52.2857	85.71428571	
Cost Variance	-11.5255	-24.9406	-23.0093	-14.28571429	
Planed Value	14.28571	15.4286	24.5714	45.71428571	
Cumulative PV	14.28571	29.7143	54.2857	100	20.34
Schedule Variance	0	-2	-2	-14.28571429	

Sprint-wise Resource Allocation & Effort

Team Composition

Scrum Master → Weight = 1.5
Senior Developer → Weight = 1.2
Junior Developer → Weight = 0.9

Effort Calculation

Effort (Man-Weeks) = Weight × Allocation (%)
Total Sprint Effort = Sum of all contributions

Sprint-wise Effort

Sprint 1 → 5.25 Man-Weeks
Sprint 2 → 5.46 Man-Weeks
Sprint 3 → 4.605 Man-Weeks
Sprint 4 → 5.025 Man-Weeks

Conversion

1 Man-Week = 5 Man-Days

Conclusion

Boosts efficiency using AI and data-driven insights.

Secures crops with real-time weather forecasts and disease alerts.

Connects farmers to share knowledge, problems, and expert solutions